

Skillr

(Freelancing Job Management System)



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Project Group Number: 73

Tutorial Group: 06

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Entities & their Attributes

- User: user_id, name, email, phone, role, rating, status
- Client: user_id, org_name, org_type
- Freelancer: user_id, exp_years, exp_level, portfolio, rating
- Admin: user_id
- Job: job_id, title, description, category, skills, base, job_status
- Bid: bid_id, price, status, time
- Contract: contract_id, start_date, end_date, final_price, contract_status
- Transaction: transaction_id, amount, type, date, payment_status
- Review: review_id, rating, comment, from

Relationship Sets & their properties

ISA Relationship

User ISA (Client, Freelancer, Admin)

- User $\rightarrow$ (1,1) specialization
- Client/Freelancer/Admin $\rightarrow$ (1,1)

Posts (Client — Job)

- Client $\rightarrow$ (0, N)
- Job $\rightarrow$ (1, 1)

One client can post many jobs

Every job **must** be posted by exactly one client

Job has total participation

Manages (Admin — Job)

- Admin $\rightarrow$ (0, N)
- Job $\rightarrow$ (1, 1)

Admin can manage many jobs

Every job must be managed by one admin

Job total participation

Bid (Freelancer — Job)

- Freelancer $\rightarrow$ (0, N)
- Job $\rightarrow$ (0, N)

Freelancer may place many bids

Job may receive many bids or none

Both partial participation

Reviews (Client — Freelancer)

- Client $\rightarrow (0, N)$
- Freelancer $\rightarrow (0, N)$

Review happens only after completion

Both can review multiple times over different contracts

Both partial participation

Assigns (Job — Contract)

- Job $\rightarrow (0, 1)$
- Contract $\rightarrow (1, 1)$

A job may or may not be assigned yet

Every contract must belong to exactly one job

Contract total participation

Signs (Freelancer — Contract)

- Freelancer $\rightarrow (0, N)$
- Contract $\rightarrow (1, 1)$

Freelancer may sign many contracts

Every contract must be signed by one freelancer

Contract total participation

Has (Contract — Transaction)

- Contract $\rightarrow (1, N)$
- Transaction $\rightarrow (1, 1)$

A contract has multiple payments (advance + final)

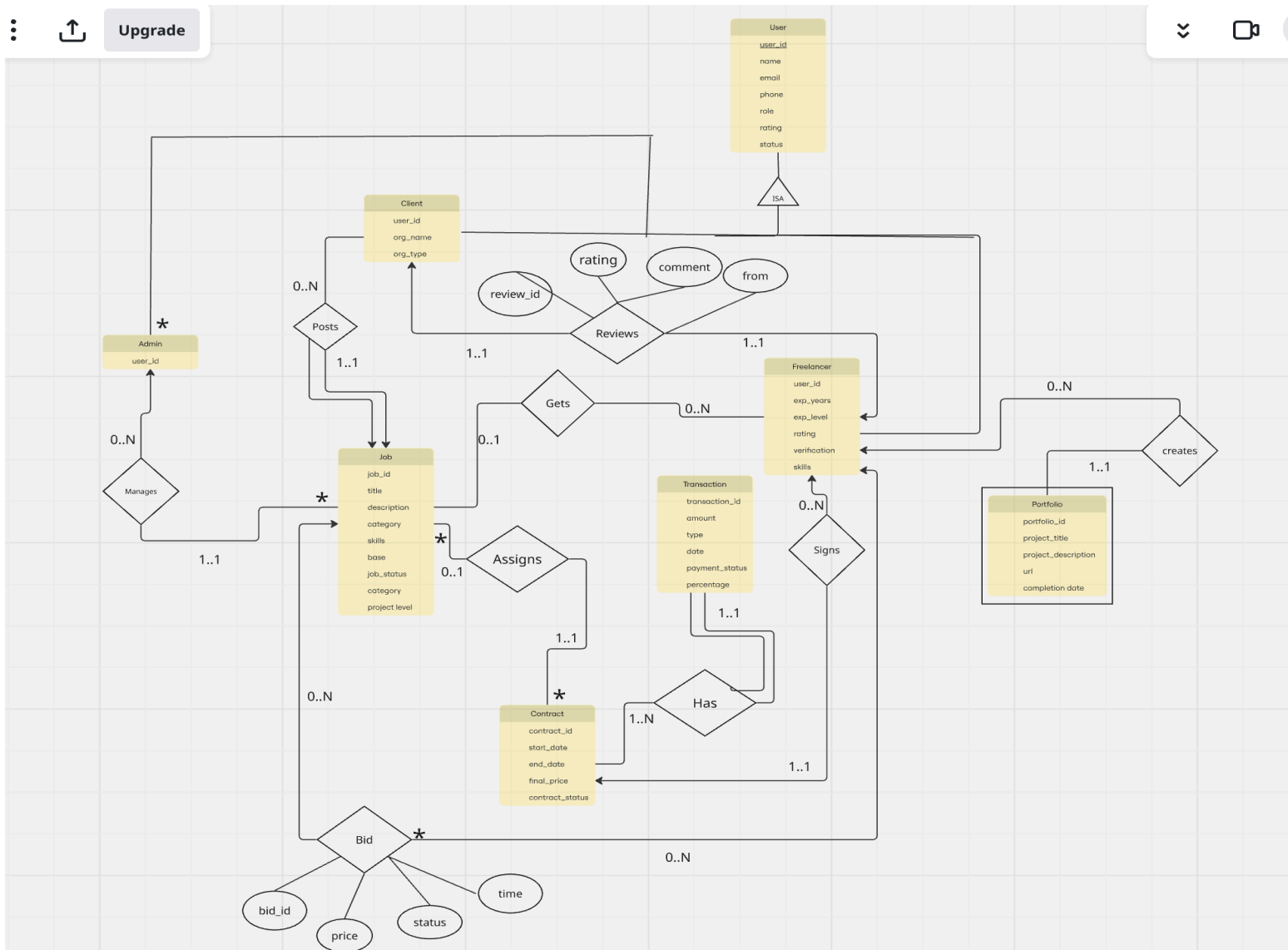
Each transaction belongs to exactly one contract

Transaction total participation

Gets (Freelancer — Job)

- Freelancer → (0, N)
 - Job → (0, 1)
- A freelancer may get many jobs
A job is finally given to at most one freelancer
Both partial participation

ER Diagram



Miro Link: https://miro.com/app/board/uXjVGMXSfrw=?share_link_id=253134177716

Relational Schemas

USER

Attribute	Key
user_id	PK
name	
email	
phone	
role	
rating	
status	

CLIENT

Attribute	Key
user_id	PK, FK(USER)
org_name	
org_type	

FREELANCER

Attribute	Key
user_id	PK, FK (USER)
exp_years	
exp_level	
portfolio	
rating	

ADMIN

Attribute	Key
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user_id	PK, FK (USER)
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JOB

Attribute	Key
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job_id	PK
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client_id	FK (CLIENT)
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title	
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description	
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category	
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skills	
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base	
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job_status	
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BID

Attribute	Key
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bid_id	PK
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job_id	FK (JOB)
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freelancer_id	FK (FREELANCER)
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price	
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time	
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status	
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CONTRACT

Attribute	Key
contract_id	PK
job_id	FK (JOB)
freelancer_id	FK (FREELANCER)
transaction_id	FK (TRANSACTION)
start_date	
end_date	
final_price	
contract_status	

TRANSACTION

Attribute	Key
transaction_id	PK
contract_id	FK(CONTRACT)
amount	
type	
date	
payment_status	

REVIEW

Attribute	Key
review_id	PK
client_id	FK (CLIENT)
freelancer_id	FK (FREELANCER)
rating	
comment	
reviewer_role	
time	

User(user_id, name, email, phone, role, rating, status)

Admin(user_id)

Client(user_id, org_name, org_type)

Freelancer(user_id, exp_years, exp_level, rating, verification, skills)

Job(job_id, title, description, category, skills, bids, job_status, category, project_level)

Portfolio(portfolio_id, project_title, project_description, url, completion_date)

Transaction(transaction_id, amount, type, date, payment_status, percentage)

Contract(contract_id, start_date, end_date, final_price, contract_status)

Manages(user_id, job_id)

Posts(user_id, job_id)

Reviews(review_id, user_id, user_id_from, rating, comment)

Gets(user_id, job_id)

Assigns(job_id, contract_id)

Has(contract_id, transaction_id)

Signs(user_id, transaction_id)

Creates(user_id, portfolio_id)

Bid(bid_id, job_id, time, status, price)

User

user-id	name	email	phone	role	rating	status
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Admin

user-id

Client

user-id	org-name	org-type
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Freelancer

user-id	exp-yr	exp-level	rating	verification	skills
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Managers

user-id	org-id
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posts/jobs

user-id	job-id
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Job

job-id	title	description	category	skills	hids	status	category	level
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Portfolio

portfolio-id	project title	project description	url	completion-date
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Transaction

transaction-id	amount	type	date	status	percentage
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Contract

contract-id	start-date	end-date	final-price	status
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Review

review-id	user-id	from	rating	comment
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Creates

user-id	portfolio-id
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Has

contract-id	transaction-id
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